

Xiu Ying Loh

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EDUCATION

Nanyang Technological University (NTU) — *Bachelor of Engineering in Computer Science* **2017 - 2021**

- Graduated with honors (Cum Laude); completed capstone project on efficient Android memory management using custom GC algorithms, published findings in university tech journal.

EXPERIENCE

Grab — *Android Developer* **Mar 2022 - Present**

- Architected and implemented a modular payment gateway abstraction layer using repository pattern, reducing transaction processing latency by 34% and enabling seamless integration of 4 new payment providers.
- Designed Kotlin Coroutines-based background synchronization system for offline-first ride history, decreasing API calls by 52% and improving app launch time from 2.8s to 1.9s for users in low-connectivity regions.
- Led code review initiatives across 8-person Android team, maintaining 98% test coverage and reducing critical bugs in production by 41% quarter-over-quarter.

ByteDance (TikTok) — *Mid-Level Android Engineer* **Jul 2021 - Feb 2022**

- Optimized video rendering pipeline using ExoPlayer 2 and hardware acceleration, reducing jank frames by 67% and achieving 60fps playback stability across 89% of Android 8+ devices.
- Refactored legacy SharedPreferences usage to encrypted DataStore, securing sensitive user data and reducing app crashes related to ANR violations by 28% within Q3.
- Collaborated with ML team to integrate on-device TensorFlow Lite model for video recommendation ranking, processing 15,000+ recommendations daily with <200ms inference latency.

Shopee — *Junior Android Developer* **Jan 2021 - Jun 2021**

- Developed real-time inventory sync feature using Firebase Realtime Database and WorkManager, handling 10,000+ concurrent product updates with 99.4% data consistency.
- Implemented custom view hierarchy for dynamic checkout UI using Jetpack Compose (pilot phase), reducing layout inflation time by 43% compared to XML-based approach.
- Fixed 127 high-priority bugs in payment flow module, improving transaction success rate from 94.2% to 98.1% across Southeast Asian markets.

PROJECTS

LocalMate - Community Task Marketplace — *Kotlin, Jetpack Compose, Firebase, Room Database, Google Maps SDK* **Aug 2022 - Dec 2022**

- Built end-to-end Android app enabling users to post and accept local tasks with real-time location tracking, attracting 8,000+ active users in Singapore and Malaysia.
- Engineered encrypted peer-to-peer messaging system using Firebase Cloud Messaging and RSA encryption, processing 45,000+ messages daily with zero data breaches.

StockNotify - Real-time Market Alert Application — *Kotlin, MVVM, Retrofit, RxJava, SQLite, WebSocket* **Feb 2022 - May 2022**

- Developed market data streaming module using WebSocket integration with Finnhub API, delivering price alerts to 12,000+ users with <100ms latency.
- Implemented sophisticated filtering and notification scheduling logic reducing battery drain by 38% through intelligent background worker optimization.

EXTRACURRICULAR

Android Developer Community Singapore — *Technical Mentor* **Jan 2022 - Present**

- Mentored 6 junior Android developers on modern architecture patterns (MVVM, MVI) and Jetpack libraries, with 5 successfully transitioning to mid-level roles within 12 months.
- Organized quarterly technical workshops covering Compose migration and performance optimization, attracting 200+ attendees across Southeast Asia.

Product Hunt — *Launch Contributor* **Jun 2022 - Present**

- Shipped 2 indie Android apps (LocalMate, StockNotify) achieving #3 and #8 rankings respectively on launch day with combined 18,000+ upvotes.

TECHNICAL SKILLS

Languages – English (Fluent), Mandarin Chinese (Native), Malay (Conversational)

Technologies – Kotlin, Java, XML, Jetpack (Compose, Navigation, ViewModel, LiveData, Room, WorkManager, DataStore), Firebase (Realtime DB, Cloud Messaging, Analytics), Retrofit, RxJava, ExoPlayer, TensorFlow Lite

Tools – Android Studio, Git, GitHub, Jira, Gradle, Firebase Console, Charles Proxy, Android Profiler, Figma